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Unit “ 1 : Session “ 8 : SLO “ 1 | Asymptotic Notations “ Big O, Big Omega, Big Theta

I. Fill in the Blanks

1. Big O “ the upper bound on running time.
2. Big Omega ($\hat{\Omega}$) “ the lower bound.
3. Big Theta ($\hat{\Theta}$) “ the tight bound.
4. worst “ Big O analyzes worst-case performance.
5. best “ Big Omega analyzes best-case performance.
6. upper and lower “ $\hat{\Theta}$ bounds from both sides.
7. $O(n)$ “ linear search.
8. $O(\log n)$ “ binary search.
9. $O(n^2)$ “ bubble sort worst case.
10. O “ $f(n) = 3n + 2$ is $O(n)$.
11. O “ $f(n) = 5n^2 + 3n + 7$ is $O(n^2)$.
12. lower-order “ constants and lower-order terms are ignored.
13. $O(1)$ (constant) “ same time regardless of input size.
14. $\hat{\Theta}$ “ $f(n) \asymp c \cdot g(n)$ for $n \asymp n_0, c$ means $f(n) = \hat{\Theta}(g(n))$.
15. O and $\hat{\Theta}$ “ $\hat{\Theta}(g(n))$ means both $O(g(n))$ and $\hat{\Omega}(g(n))$.